

0	4	.	1
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Describe and explain the role of antibodies in stimulating phagocytosis.

Do **not** include details about the process of phagocytosis.

[2 marks]

Question 4 continues on the next page

Turn over ►



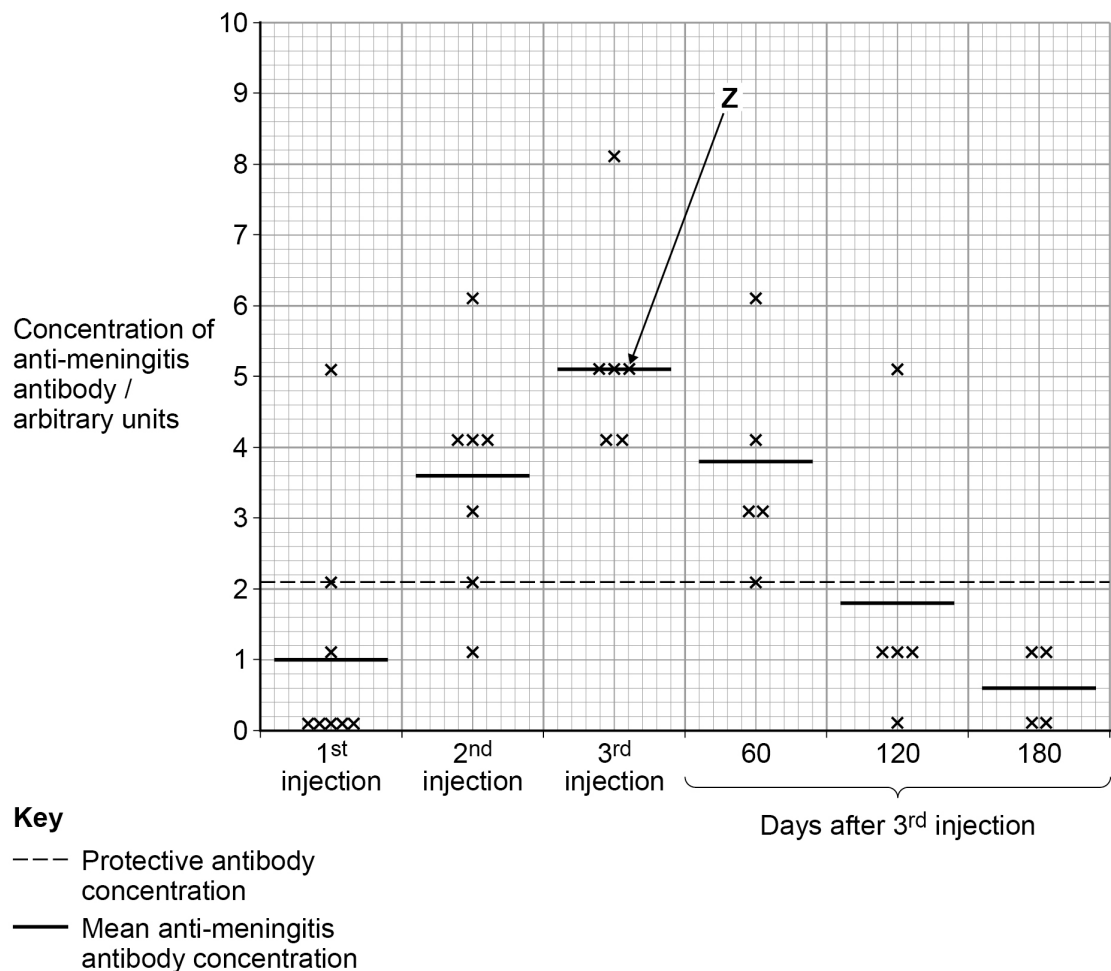
Meningococcus bacteria cause a disease called meningitis. Scientists investigated a new meningitis vaccine (MenG) by measuring changes in blood anti-meningitis antibody concentration in mice.

Each mouse was given three separate MenG injections. The concentration of anti-meningitis antibody was measured in a sample of blood taken soon after each injection.

After the 3rd injection, the concentration of anti-meningitis antibody in the blood was also measured after 60 days, after 120 days and then after 180 days.

Figure 4 shows the scientists' results. Each plotted point in **Figure 4** is the result for a different mouse.

Figure 4



Do not write
outside the
box

0	4	.	2
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The scientists discovered that the concentration of anti-meningitis antibody of the mouse labelled **Z** in **Figure 4** decreased after the 3rd injection at a constant rate of 0.027 arbitrary units per day.

Use this information and **Figure 4** to calculate the number of days after the 3rd injection the antibody concentration is higher than the protective antibody concentration for this mouse.

[2 marks]

Answer = days

0	4	.	3
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Using **Figure 4**, what can you conclude about the effectiveness of each injection on the immune response of these mice?

[4 marks]

[illegible]

Turn over ►



0	4	.	4
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The scientists hypothesised that memory B cells had formed in the mice 180 days after the 3rd injection.

Suggest and explain a practical method the scientists could use to test this hypothesis.

[2 marks]

[Extra space]

10

